SPACE SYSTEMS

CATALOGUE

A Division of Electro Optic Systems Pty Limited

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2023 CATALOGUE

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EOS Space Systems are a global leader in the design, manufacture, delivery and operation of sensors and systems for space domain awareness and space control.

CONTENTS

SPACE INTELLIGENCE FOR BATTLEFIELD COMMANDERS

ACHIEVING MISSION SUCCESS BY DENYING ENEMY INTELLIGENCE TO THE STRATEGIC HIGH GROUND OF SPACE

Maintaining the element of surprise is a key factor on the modern battlefield. Knowledge of when, where and for how long enemy Earth observing satellites are overhead will give allied forces a distinct military advantage.

Today Electro Optic Systems (EOS) is using Australian-developed technology to produce highly accurate real-time data and intelligence products which leverage EOS' catalogue of Objects of Interest to Australian and allied defence clients.

This information is monitored daily using EOS' Australian network of ground-based sensors, which can track thousands of objects per day for any potential changes in pattern-of-life and threats to operational satellites.

Battlefield commanders can use this real-time data to conduct safe mission planning to ensure operations are protected from enemy surveillance, thereby enabling the highest probability for mission success.

SPACE DOMAIN AWARENESS

EOS has been a proud partner of the Australian Department of Defence in Space Domain Awareness (SDA) for more than 15 years. Through a proven network of passive and active optical SDA sensors, EOS can cover all orbit regimes from LEO to GEO with unrivalled accuracy. Having highly accurate SDA data enables better intelligence on knowing when, where and who is watching at any point in time.

SDA SERVICES

- Live threat warning for commanders before and during the mission, supporting route planning and updated notifications based on live orbital manoeuvres
- Regular space intelligence summaries
- Independent and Australian space catalogue
- Object characterisation and identification services
- Data analytics, machine learning and artificial intelligence
- Simulation tools for mission planning and capability assessments

DATA SERVICES

- Accurate tracking from LEO to Cis-Lunar
- Taskable network of sensors for direct tasking based on mission needs
- 24 hour coverage
- Region search, object detection and tracking
- Satellite laser ranging
- High-rate image capture

ACCURATE

EOS sensors provide unrivalled positioning accuracy of small, dim and far objects for SDA and space control during the day and night.

PROVEN

EOS' SDA services are proven in global space operations for military and civilian applications. With high fidelity tracking, actions in space are captured in detail.

MISSION READY

EOS' network of taskable sensors are interoperable with other networks and operationally tested and ready for data delivery.

SPACE INTELLIGENCE AND CONTROL

ENSURING FREEDOM OF ACTION AND ACCESS TO THE SPACE DOMAIN

EOS is Australia's trusted and proven leader in delivering unique optical surveillance capabilities for space domain awareness (SDA), space intelligence, and space control.

EOS provides engineering services in the defence, government, and commercial sectors specialising in cutting-edge directed energy capabilities for space object tracking, characterisation, identification, optical communications, remote manoeuvre and missile defence. With in-house design and development expertise, EOS develops new products to meet the changing strategic needs of Australia and allied nations.

SDA SERVICES

- Live threat warning for commanders before and during the mission, supporting route planning and updated notifications based on live orbital manoeuvres
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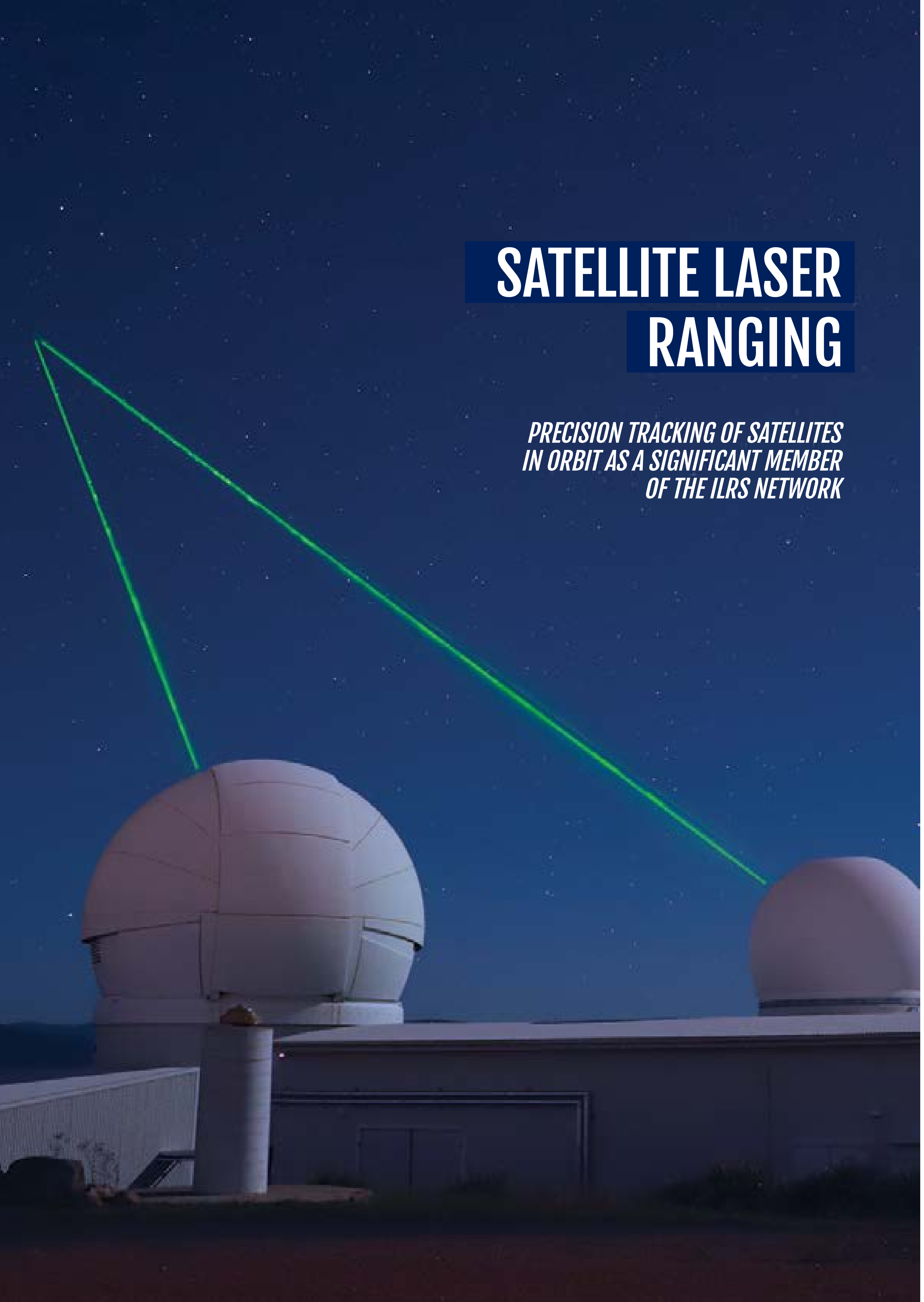


SATELLITE LASER RANGING

PRECISION TRACKING OF SATELLITES IN ORBIT AS A SIGNIFICANT MEMBER OF THE ILRS NETWORK

Electro Optic Systems is a world-leader in the design, construction and operation of satellite laser ranging (SLR) stations.

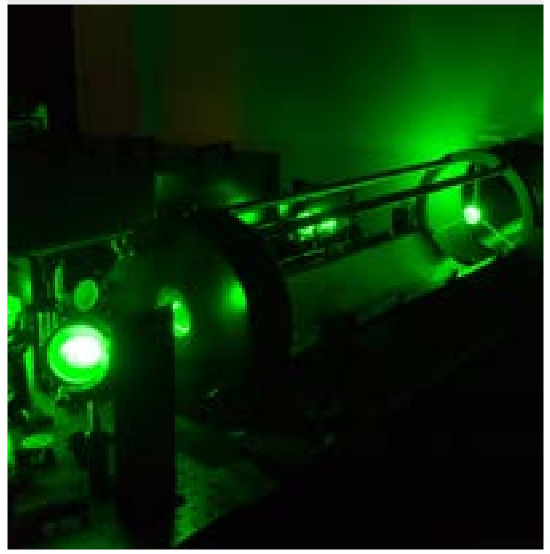
Satellite laser ranging accurately tracks satellites in orbit with exceptional precision, measuring movements at the millimetre level. This is crucial for accurately determining satellite orbits, which is essential for geodesy, environmental monitoring, and establishing global coordinate systems such as that used by GPS.

EOS' SLR stations contain state-of-the-art lasers, timing systems, beam directors and fully automated software, all developed in-house. Our end-to-end approach to the SLR station development, combined with our extensive domain expertise, positions EOS' SLR products among the highest performing worldwide.

As a prominent member of the International Laser Ranging Services (ILRS) network, we have been providing data to Geoscience Australia since 2004 from our Mt Stromlo facility in Canberra. EOS has gained a reputation for providing turnkey SLR stations of the highest quality, successfully deploying these stations to clients in Europe and Asia.

FEATURES

- Automated and remote controlled
- Ranging to high and low satellites with few-millimetres resolution
- Network synchronised tracking
- Standard systems of 1 m aperture
- Weatherproof in operation



PRECISION TRACKING

EOS' SLR stations provide unrivalled positioning accuracy of satellite orbits, capable of measuring movements down to the millimetre.

HIGH PERFORMANCE

EOS' SLR stations contain state-of-the-art laser technology and fully-automated functionality for high performing turnkey solutions.

NETWORK SOLUTIONS

EOS have been providing accurate satellite tracking data to the ILRS network since 2004.

SPACE SUSTAINABILITY

SPACE DEBRIS MITIGATION THROUGH HIGH ENERGY LASER TECHNOLOGY

The risk of a catastrophic collision in space between active satellites and debris is rapidly escalating, driven by the increased rate of satellite launches resulting from reduced space access costs.

The creation and management of space debris pose a globally recognised problem with far-reaching consequences, potentially impeding future access to space for generations to come, especially in Low Earth Orbit (LEO).

EOS has a long-standing program dedicated to space debris mitigation, employing laser technology. We are actively seeking investment partners, clients, and collaborators to develop and deploy this capability. Our approach to space debris mitigation involves utilising high energy lasers to illuminate debris, altering its original orbit and preventing potential collisions with valuable space assets.

SPACE DEBRIS MITIGATION

- Ground-based laser debris collision avoidance system
- Reduces debris-to-satellite collisions
- Cost-effective, does not require a launch
- Agile, beam director can rapidly move from one object to the next
- Low maintenance and longer lifetime than space-based solutions
- Assists active debris removal missions
- Safe laser manoeuvre operations based on EOS' safety of flight collision analysis tools

COLLABORATIVE

EOS partners with other networks to monitor space debris and protect vital assets.

ACCESS TO SPACE

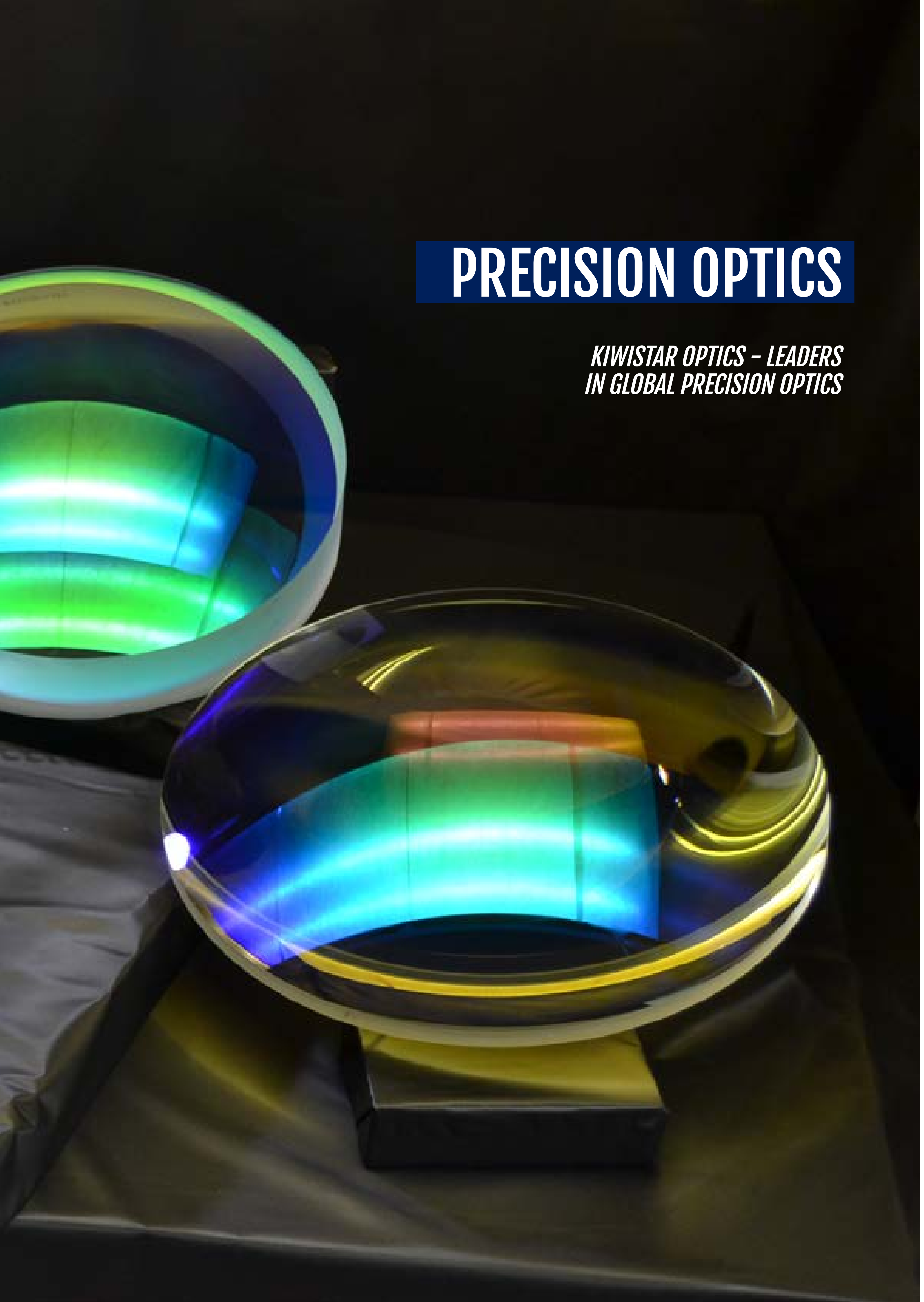
Space sustainability improves future generations access to space and reduces global risks to valuable infrastructure.

MISSION READY

EOS' network of taskable sensors are interoperable with other networks and operationally tested and ready for data delivery.

PRECISION OPTICS

KIWISTAR OPTICS – LEADERS IN GLOBAL PRECISION OPTICS



KiwiStar Optics designs, manufactures, and delivers everything from one-off optical elements to complete optical-mechanical systems.

KiwiStar Optics has been manufacturing high precision optics since 1947. We've made some of the planet's biggest lenses for its best telescopes. Our reputation as a global leader in precision optics is built on consistent excellence and innovation.

RELIABILITY AT THE CUTTING-EDGE OF OPTICAL FABRICATION

We provide a full range of services which includes system design and analysis, manufacture, assembly and integration, metrology, testing and quality assurance to ensure high performing optics, instrumentation and optical assemblies.

Our manufacturing capability for all optics is extensive. We specialise in "metre-class" optics with capacity for up to 2 m diameter aspherical lenses or mirrors; and can apply thin-film multilayer coating up to 850 mm diameter.

KiwiStar's reputation as a global leader in precision optics is built on consistent delivery of the most challenging optical products. Our scientists, engineers and master opticians draw on decades of experience and a passion for what they do.

KiwiStar Optics is part of the EOS Space Systems division and is a wholly owned subsidiary of Electro Optic Systems Pty Limited.

Visit www.kiwistaroptics.com

CUSTOMER FOCUSED

Our international track record is built on successfully delivering to our customers on time, to specification and in budget.

RELIABLE

Consistent excellence is our aim, and our international track record of successfully completed projects is testimony to that consistency.

COMPETITIVE

Our facilities, processes, equipment and people are world-class.

Our results are proven. We can point to a succession of wide field correctors, optical components, cameras, and other instruments we have completed for many of the leading observatories around the world.

INNOVATIVE

We invest in technology development to supply innovative solutions for our clients.

DESIGN

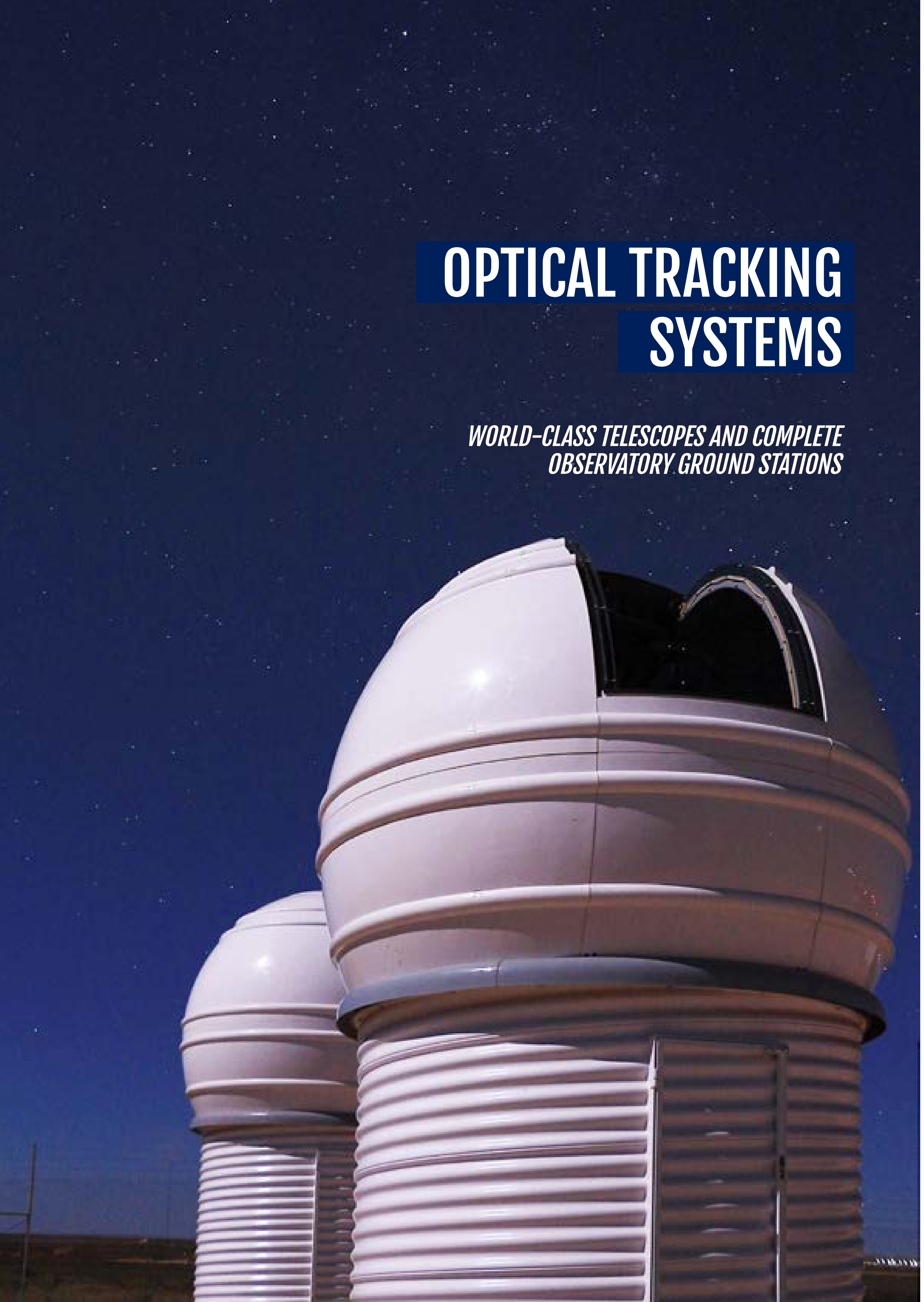
World-class design and development of optics and opto-mechanical components and assemblies.

MANUFACTURE

Precision optical-mechanical manufacturing facilities matched by high precision testing, metrology and quality systems.

INTEGRATION

Experience counts when integrating and tightly aligning tolerance optics with mechanical components for instrumentation and optical assemblies.



OPTICAL TRACKING SYSTEMS

WORLD-CLASS TELESCOPES AND COMPLETE OBSERVATORY GROUND STATIONS

EOS delivers precision electro-optic solutions for high performance day and night space object tracking, specialising in laser ranging.

We provide leading technology which can accurately track satellites, debris, and other objects in all orbital regimes – LEO, MEO, GEO and cislunar.

Our product range covers complete observatory and mission control facilities with automated and remote operations, to individual sub-systems such as high accuracy laser beam directors, high power laser systems, mission systems, imaging systems, space network management software, and timing systems.

EOS has a proven track-record in designing, installing, commissioning and maintaining complete observatory facilities worldwide. Our clients in North America, Europe and Asia-Pacific rely on our expertise. These facilities are low-maintenance and offer long-term reliability.

TELESCOPES

World-class high-speed alt-azimuth telescopes for imaging, laser beam direction and tracking.

OBSERVATORIES

Complete observatory ground stations, including design, build, installation, commissioning, maintenance, and sustainment.

TELESCOPES

- 0.7 m to 2.4 m aperture
- 0.7 m transportable and complaint with MIL-810H (vibe, shock, environment)
- Slew rate: better than 1 rad/sec
- Pointing accuracy: better than 2.5 arcsec
- Pointing precision: better than 0.1 arcsec at sidereal velocity
- Tracking accuracy: better than 1 arcsec over a 10-minute period at sidereal velocity

OBSERVATORIES

- Software controls for automated or remote-controlled operation of facilities, including automated weather close
- Telescope apertures from 0.7 m to 2.5 m
- Passive tracking systems (visible and infrared)
- Laser tracking systems (satellite and debris)
- Domes, including long time between failure of drive mechanisms and all-weather sealed enclosures
- Off-grid operations

WORLD-CLASS LEADERS

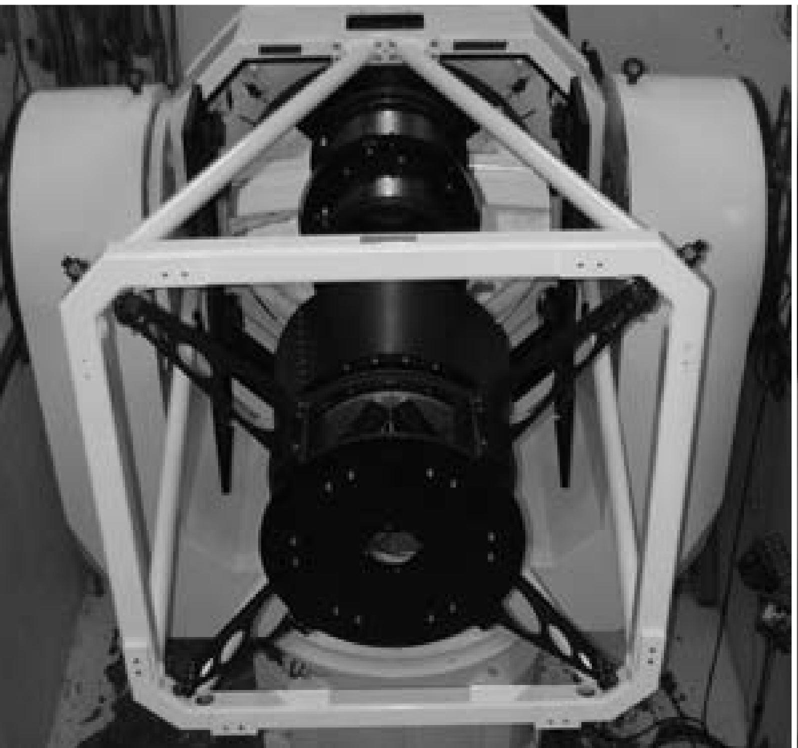
EOS' observatory ground stations and telescopes rank in the world's top 10 for high performance space object tracking.

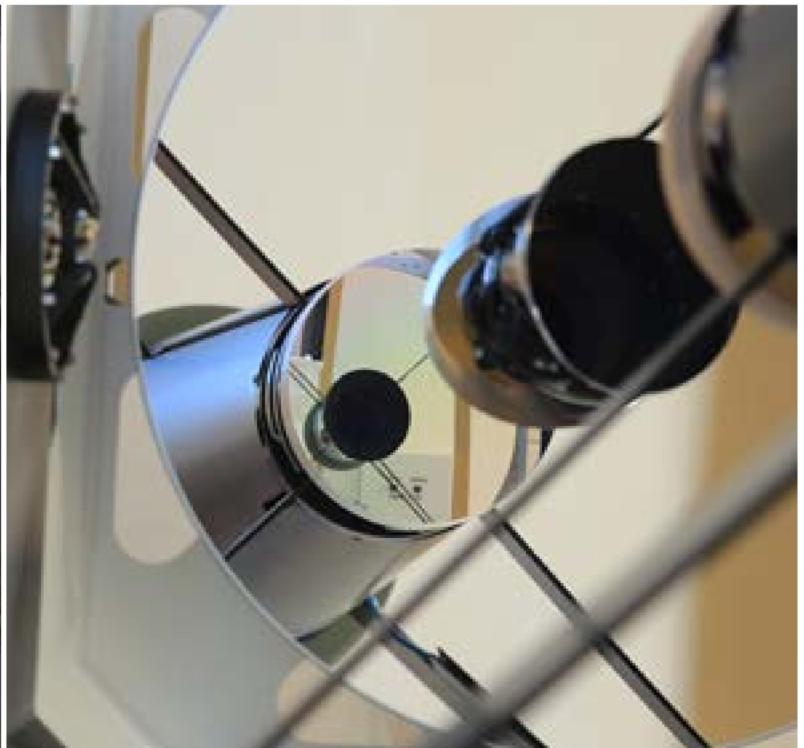
PROVEN TRACK RECORD

EOS design, install, commission and maintain complete observatory systems world-wide. From laser systems to fully-automated remote operations.

SOFTWARE CONTROLS

Automated and remote-controlled operations available which offer long-term reliability and regular updates.



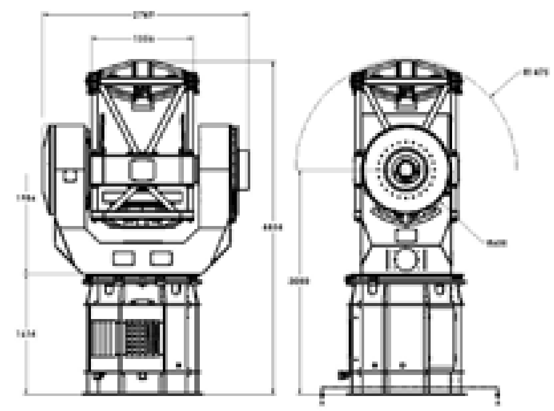


1 METRE CLASS TELESCOPE

EOS designs and manufactures high-quality telescopes with exceptional technical performance for imaging, tracking and laser beam delivery.

Our telescopes are agile, highly accurate, reliable, robust and low maintenance. They seamlessly integrate with telescope enclosures, instruments, and control software. Equipped with remote diagnostic support and automated operation, and are easily upgraded. Used as laser beam directors, they provide the accuracy required for reliable ground-to-space optical communications and satellite laser ranging.

The 1 m telescope is compact and high-speed, fits domes larger than 4.5 m and can provide a large-diameter Coudé path. It is designed for high positioning accuracy and ultra-smooth tracking with large instrument payloads.



1 Metre Class can be used for apertures up to 1.4 m

FEATURES

Telescope configuration	Nasmyth and Coudé	
Aperture size	0.7–1.2 m	
Primary mirror focal ratios	• From f/1.5 to f/1.8 • Optical and IR optimised designs	• Passive mirror supports
System focal ratios	From f/8 to Mersenne	
Mount	Compact alt-azimuth	
Axis speeds	• Axis speeds > 10 deg/s	• On-axis direct drives and absolute encoders
Tracking jitter	< 0.25 arcsec RMS	
Pointing accuracy	< 3 arcsec RMS	
Operational features	• Wide field guide scopes • Active collimation • Remote autonomous operation • Integrated dome and observatory systems • Wide field corrector systems	• Fast tip-tilt secondary mirrors • Instrument derotators • On-board instrument processors • Automatic sky calibration software • Turnkey remote control observatories

HIGH RELIABILITY

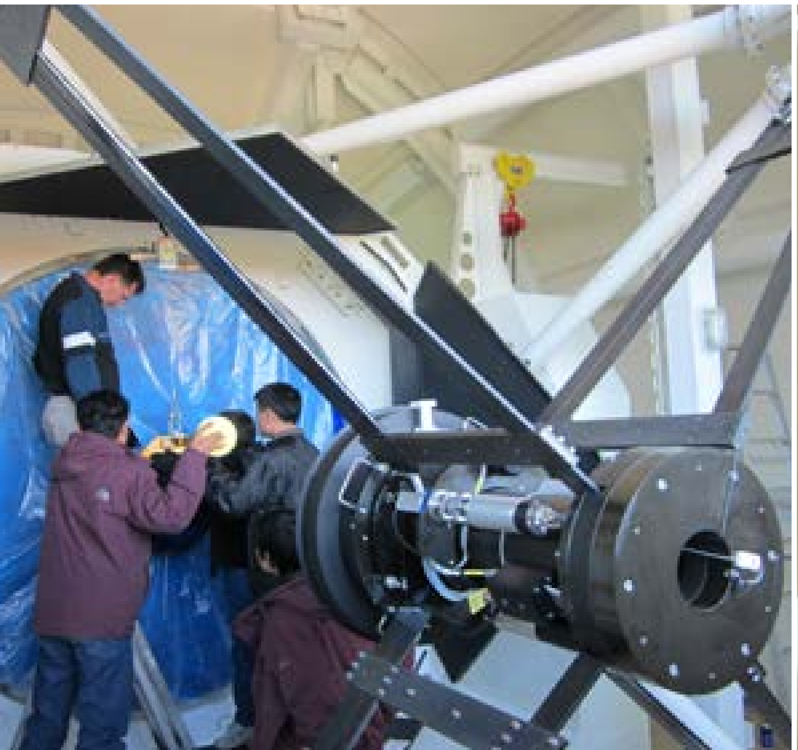
Our telescopes are robust, reliable and low maintenance.

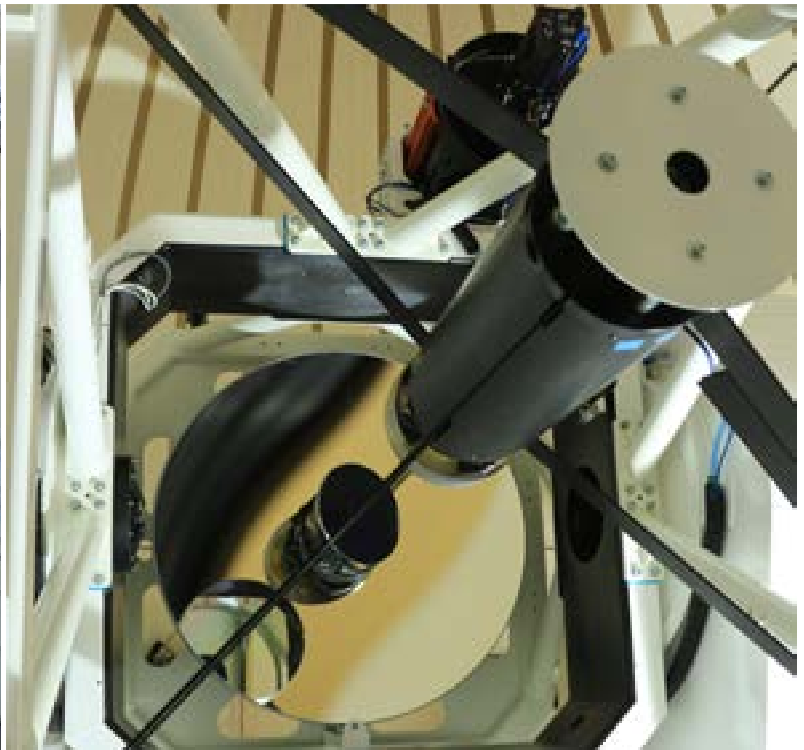
ULTRA-SMOOTH TRACKING

Designed for high positioning accuracy with large payloads.

EXCEPTIONAL PERFORMANCE

Optimised for ten-year mean time between failure.



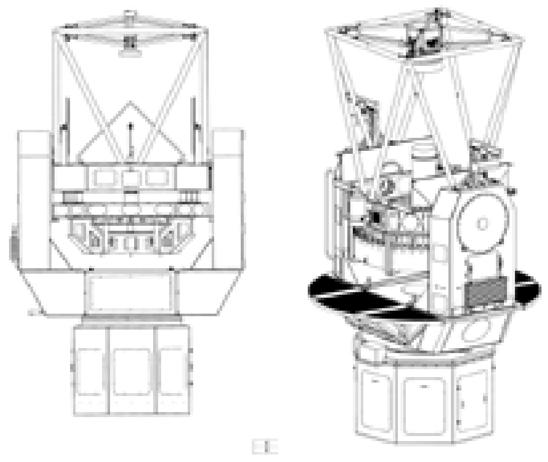


2 METRE CLASS TELESCOPE

EOS designs and manufactures high-quality telescopes with exceptional technical performance for imaging, tracking and laser beam delivery.

Our telescopes are agile, highly accurate, reliable, robust and low maintenance. They seamlessly integrate with telescope enclosures, instruments, and control software. Equipped with remote diagnostic support and automated operation and are easily upgraded. Used as laser beam directors, they provide the accuracy required for reliable ground-to-space optical communications and satellite laser ranging.

The 2.4 m telescope is ideally suited to astronomy where reliability, ease of maintenance and high image quality are key. It has superb performance in slewing and tracking and can be built with Cassegrain or Nasmyth foci, making it a versatile telescope to support diverse astronomical research.



2 Metre Class can be used for apertures from 1.8m to 2.4m

FEATURES

Telescope configuration	Nasmyth, Coudé or Cassegrain	
Aperture size	1.8–2.4 m	
Primary mirror focal ratios	• From f/1.5 to f/2 • Optical and IR optimised designs	• Passive mirror supports
System focal ratios	From f/8 to Mersenne	
Mount	Compact alt-azimuth	
Axis speeds	• Axis speeds > 10 deg/s	• On-axis direct drives and absolute encoders
Tracking jitter	< 0.5 arcsec RMS	
Pointing accuracy	< 3 arcsec RMS	
Operational features	• Active collimation • Remote autonomous operation • Integrated dome and observatory systems • Wide field corrector systems	• Fast tip-tilt secondary mirrors • Instrument derotators • Complete remote control observatories

HIGH RELIABILITY

Our telescopes are robust, reliable and low maintenance.

ULTRA-SMOOTH TRACKING

Designed for high positioning accuracy with large payloads.

EXCEPTIONAL PERFORMANCE

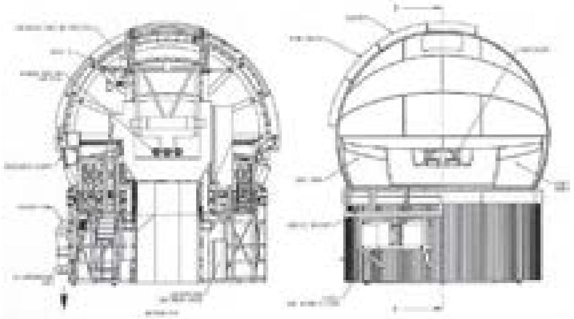
Optimised for ten-year mean time between failure.

ICESTORM II ENCLOSURE

EOS provides turnkey observatory stations that are highly automated for 24/7 unstaffed operations, even in harsh environment conditions.

We are experts in designing, installing, commissioning, operating and maintaining best-in-class observatories. This allows us to offer unique solutions that meet the most challenging customer demands.

The IceStorm II Enclosure is suited to large beam directors, and accommodates apertures up to 2.4 m; this makes it ideal for optical astronomy applications. Its robust dome construction minimises thermal effects, protects against vibration and reduces wind shake.



Dome Schematic showing the three floors.

FEATURES

Structure Capable for remote operations	• 8 m diameter ringwall supporting a co-rotating spherical enclosure with diameter of 9 m • Composite GRP dome panels includes fire retardant insulation (>R3 or R19 US); dual flap seals minimise air leakage
Aperture size	1.8–2.4 m
Dome rotation	• Accurately track telescope elevation at speeds up to 2°/sec • Dual tracking (positioning to < 50mm) • Speed: 2°/sec and acceleration 1°/sec²
Dome acceleration	• Co-rotating dome maximises space, improves safety and minimises volume for environmental conditioning • Dome rotational speed: 4°/sec
Operational environment	> 1°/sec ²
Operational features	• Level 4 mezzanine floor • Service crane • HVAC • Nasmyth instrument hatch • Instrument cooling • Lightning protection

HIGHLY AUTOMATED

Capable of 24/7 remote operations.

QUALITY COMPLIANT

Certified ISO 9001.
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ACCURATE TRACKING

Accurately track telescope elevation up to 2°/sec.

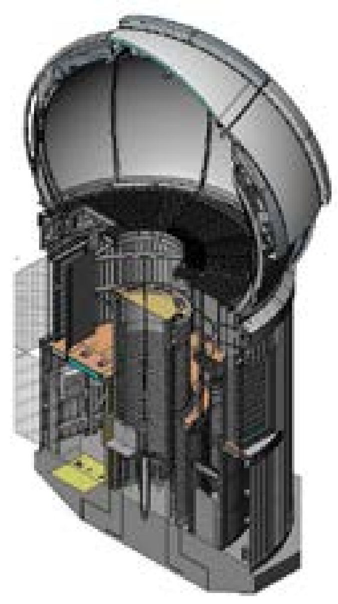
TEMPEST ENCLOSURE

EOS provides turnkey observatory stations that are highly automated for 24/7 unstaffed operations, even in harsh environment conditions.

We are experts in designing, installing, commissioning, operating and maintaining best-in-class observatories. This allows us to offer unique solutions that meet the most challenging customer demands.

The Tempest Enclosure is optimised in design for beam directors, and optical communications and is ideal for astronomy applications. It incorporates a composite glass-reinforced plastic (GRP) rotating dome with vertical (up-and-over) bi-parting shutters.

The pre-assembled modular dome and ringwall kit minimises on-site construction and integration time. This proven design also reduces ongoing maintenance costs, making it an economical system to run.



Tempest observatory system with standard EOS design and supply options.

FEATURES

Structure	Three-level ringwall: • Level 1: insulated equipment floor • Level 2: ventilation floor with four vent doors • Level 3: fixed observing floor with access hatch	
Aperture configuration	Up to and including 1.5 m in Cassegrain, Nasmyth and Coudé	
Diameter	6.5 m diameter standard, with 1.75 m slit	
Shutters (providing protection against wind, shake and stray light)	• Dual tracking (positioning to < 50 mm) • Speed: 2°/sec and acceleration 1°/sec²	
Azimuth speed	4°/sec and acceleration 1°/sec²	
Viewing	Unvignetted viewing from 10 – 90° in elevation	
Operating environment	• Operational wind load to 80 km/h • Fire retardant insulation (>R3 or R19 US), ventilation floor, active cooling system, thermally isolated area for equipment racks	
Optional features	• Pier forms • Tower frame • Ringwall panels	• Lightning protection • HVAC

HIGHLY AUTOMATED

Capable of 24/7 remote operations.

QUALITY COMPLIANT

Certified ISO 9001.

ACCURATE TRACKING

Accurately track telescope elevation up to 2°/sec.

TYPHOON ENCLOSURE

EOS provides turnkey observatory stations that are highly automated for 24/7 unstaffed operations, even in harsh environment conditions.

We are experts in designing, installing, commissioning, operating and maintaining best-in-class observatories. This allows us to offer unique solutions that meet the most challenging customer demands.

The Typhoon Enclosure is a two-axis, all-weather dome for unattended operations in harsh environments, allowing for full thermal control of the telescope inside. The dome uses an extended hemisphere construction to provide unprecedented strength, and is built using high-strength, fibre-reinforced polymers.

Typhoon observatory system is fully enclosed.

APPLICATIONS

- Astronomy
- Optical tracking
- Satellite laser ranging
- Laser communications

FEATURES

Structure	• All-weather and enclosed • Suitable for remote, unmanned locations in harsh environments • Standard and custom prefabricated towers • Two axis with multiple windows and/or apertures • Access stairs	
Diameter	4.5 m diameter standard; custom designs available	
Windows	• Standard window positioning accelerations to 20°/sec² and speeds to 20°/sec	• Specifications are customisable
Operating environment	• High strength for 200 km/h wind loads • High thermal insulation for precise seeing control and telescope thermal conditioning • Optional seals for pressurised, dustproof operation to cleanroom specifications	• Telescopes operate in an ideal, clean, controlled environment extending telescope MTBF (MTBF > 60,000 operational hours when properly maintained)

HIGHLY AUTOMATED

Capable of 24/7 remote operations.

QUALITY COMPLIANT

Certified ISO 9001.

ACCURATE TRACKING

Accurately track telescope elevation up to 2°/sec.

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